Xuanyou Chen

960 Spring Street Northwest, Atlanta, GA 30309 | (760) 390-0895 | xchen3019@gatech.edu | Personal Website

Education

Georgia Institute of Technology | Atlanta, GA Aug 2025 - May 2027
Bachelor of Science in Computer Engineering GPA: **4.0/4.0**
Specializations: Distributed Systems & Software Design; Robotics & Autonomous Systems

Emory University | Atlanta, GA Aug 2022 - May 2025
Bachelor of Science in Computer Science and Mathematics GPA: **4.0/4.0**

Research

VerifLM: Physically Verified Goal Pose Generation for Manipulation *(First Author, Under Review at ICRA 2027)* Jan 2026 - Sep 2026
Trustworthy Robotics Lab, Georgia Institute of Technology

- Built a VLM-based task planner converting human instructions and RGB-D images into physically verified 6D object goal poses.
- Developed a perception pipeline using VGGT for depth estimation, GroundingDINO and SAM2 for object detection and segmentation, FoundationPose for 6-DoF pose estimation, and Hunyuan3D for online mesh reconstruction.
- Designed a novel VLM reasoning interface that identifies incorrect translational and rotational components, refines candidate object goal poses, and physically validates them in MuJoCo.
- Utilized AnyGrasp and OMPL RRT-Connect to generate grasps and collision-free trajectories, executed through ROS 2 and Drake.
- Achieved 74.07% task-planning and 71.76% real-robot execution success across nine manipulation tasks (160/216 and 155/216 trials), outperforming OmniManip (43.98%) and ReKep (32.87%) in task planning; parallel-world validation improved mesh-free execution from 42.59% to 60.19% (92/216 to 130/216).

MyCobot280 Manipulator Infrastructure Development Aug 2025 - May 2026
Intelligent Vision and Automation Lab, Georgia Institute of Technology

- Built a Python package supporting kinematics, gripper control, trajectory recording for the MyCobot280 arm and camera calibration and image capture for Intel RealSense D435, OAK-D, and Microsoft Kinect v1.
- Collaborated on ArUco-based camera calibration and implemented a "click-to-capture" vision-guided picking pipeline.
- Customized the full hardware stack for a suction gripper, including a MOSFET-based pump/valve switching circuit, GPIO interfaces, and 3D-printed mechanical adapter for mounting onto the arm.
- Diagnosed and repaired two malfunctioning electric grippers—opened and serviced the hardware, coordinated with vendor technical support to secure replacement servos, restoring both grippers to working condition.

End-to-End Imitation Learning on LeRobot SO-101 | Project Webpage Nov 2025 - Jun 2026
Tools: LeRobot SO-101, PyTorch, ACT, ResNet-18

- Built a dual-arm teleop system and an imitation-learning policy mapping RGB images and proprioception to joint commands.
- Trained policies across three datasets: 10 demos/5,557 samples, 50 demos/9,600 samples, and 100 demos/70,904 samples.
- Implemented an ACT-style policy with a pretrained ResNet-18 encoder and 4-layer Transformer to predict 30-step action chunks; deployed with receding-horizon inference every three control steps.
- Achieved bottle pick-and-place and regrasp recovery after addressing insufficient data, shadow-induced spurious correlations, poor camera viewpoint, and lack of proprioceptive feedback.

Publication

[1] **Xuanyou Chen**, Jinyu Chen, Glen Chou, **"VerifLM: Physically Verified Image-Edit-Guided Goal Pose Generation for Robotic Manipulation"**, Submitted to IEEE ICRA 2027.

Work Experience

Guangzhou Pinyi Information Technology | ML Engineering Intern Jun 2024 - Aug 2024

- Contributed to a Python ML backend enabling clinical researchers and pharmacists to configure predictive models through a graphical interface, from hospital datasets to new-patient predictions.
- Developed configuration-driven training workflows integrating 15 scikit-learn estimators (8 classifiers and 7 regressors), with user-selected features, prediction targets, train/test splits, and GridSearchCV tuning for classification.
- Connected SQL Server datasets to pandas workflows and computed Pearson, Spearman, and Cramer's V correlations to support clinical feature selection.
- Supported model comparison through classification reports, confusion matrices, and regression metrics; persisted models with joblib and returned patient predictions and class probabilities as JSON.

Curastone | Software Developer Intern Sep 2023 - Dec 2023

- Assisted with the development of a Next.js AI learning assistant generating flashcards and personalized problem sets.
- Implemented user authentication, file upload, and course management, integrated with backend API using Redux.
- Designed responsive webpages using Tailwind CSS to ensure proper display of elements on various screen sizes.
- Deployed website using Vercel and AWS Route 53 and documented the deployment process for future reference.

Other Experience

RoboNav, RoboJackets | Software Team Member Sep 2025 - Dec 2025

- Implemented camera-based obstacle detection using HSV filtering to generate a top-down occupancy grid in ROS2 and C++.

- Built key components of a SLAM pipeline: implemented an odometry sensor model and particle-filter localization, and a mapping node that uses TF transforms to fuse obstacle detections into a global occupancy grid.
- Developed a hill-climbing optimization module that samples neighboring terrain and steers toward max-elevation goal points.

Emory Center for AI Learning | Project Leader Jan 2025 - May 2025

- Led the development of a medical-device support chatbot for MedView, utilizing AI to answer device-related questions.
- Developed a React & TypeScript frontend supporting both text and voice interaction using the Web Speech API.
- Built a FastAPI backend to query the DeepSeek API, with semantic caching using Sentence Transformers and FAISS for low-latency FAQ retrieval from a predefined MongoDB database.

Applied AI Ventures Lab | Research Assistant Jan 2024 - May 2024

- Developed a full-stack video analytics platform with Next.js frontend and a Spring Boot backend (42+ REST APIs).
- Designed a database schema in MySQL and deployed backend server on Linux-based AWS EC2 instances, automated data transfer from local development to Linux server using Bash scripts.
- Implemented a video content similarity model using GPT-4 & spaCy for transcript-based semantic analysis.

Emory Robotics Club | Programming Team Member Sep 2024 - Dec 2024

- Built and programmed the VEX V5 Classroom Starter Kit robot as a foundation for autonomous and driver control;
- Developed motion code using the PROS C++ library for the 2025 VEX “High Stakes” competition;
- Implemented path-planning logic and tuned PID controllers for drivetrain and lift stability.

Projects

Perception-Based Grasping and Collision-Aware Manipulation Aug 2026 - Oct 2026

Tools: Python, NumPy, SciPy, MuJoCo, AgileX PiPER

- Implemented a 3D perception pipeline using voxel downsampling, Kabsch alignment, and trimmed point-to-point ICP to estimate object poses from partial point clouds.
- Developed an analytical 6-DoF geometric Jacobian and damped-least-squares inverse kinematics solver.
- Implemented antipodal grasp generation using surface normals, friction-cone constraints, and grasp-frame construction to identify feasible parallel-jaw grasps.
- Built a bidirectional RRT-Connect motion planner with joint-space sampling, collision-checked extensions, tree connection, and path shortcutting for obstacle-aware pick-and-place.

Parallel N-Body Gravity Simulator Sep 2026 - Oct 2026

Tools: C++, OpenMP, OpenGL, CUDA

- Developed a 3D gravitational N-body simulator using a symplectic leapfrog integrator to model particle dynamics with all-pairs gravitational interactions.
- Implemented CUDA kernels to parallelize the $O(N^2)$ gravitational force computation on the GPU, alongside an OpenMP-based CPU parallel implementation.
- Built an interactive OpenGL visualization for real-time particle rendering, camera navigation, and simulation controls, separating physics computation from rendering.

Real-Time Collision Prediction for Autonomous Vehicles Jul 2025 - Sep 2025

Tools: Python, PyTorch, MuJoCo

- Reimplemented the collision probability prediction (CPP) model from Zhang et al. (TNNLS 2021) to estimate imminent collision risk from LiDAR-style range measurements and vehicle velocity states.
- Built a custom MuJoCo simulation environment and data collection pipeline, generating 272,000 labeled samples with 14-dimensional observations.
- Trained a 3-layer PyTorch MLP using weighted binary cross-entropy, achieving 99.79% validation recall and 0.4935 F1 score; integrated real-time collision predictions into the MuJoCo simulation.

Relevant Coursework

Advanced Robotic Manipulation: Point-cloud object tracking and ICP, grasp synthesis, motion planning, manipulation control, behavior cloning, with hands-on projects on the AgileX PiPER 6-DOF manipulator.
Introduction to Automation & Robotics: Forward and inverse kinematics, manipulator Jacobians, robot dynamics, trajectory planning, position/torque control, and robotic sensing, with projects on the LeRobot SO-101 arm.
Introduction to Perception & Robotics: Markov and Monte Carlo localization, particle filters, differential-drive robotics, computer vision, ICP/SLAM, and RRT-based motion planning.
Control System Design: State-space control, state estimation, tracking, discrete-time control, and motion-control systems, with embedded implementation of digital control algorithms on a TI C2000 microcontroller.
Feedback Control Systems: Dynamic system modeling, Laplace transforms, transfer functions, stability, root locus, and frequency-response design.

Skills

Programming Languages: Python, C++, C, MATLAB, Java, JavaScript/TypeScript, SQL
Robotics & Motion Planning: ROS 2, MuJoCo, Drake, OMPL, MoveIt, LeRobot
Machine Learning & Perception: PyTorch, OpenCV, VGGT, SAM2, GroundingDINO, FoundationPose, AnyGrasp, Hunyuan3D
Software Engineering: Linux, Git, React, MySQL, MongoDB, AWS, Google Cloud
Hardware & Prototyping: KUKA LBR iiwa 7 R800, AgileX PiPER, LeRobot SO-101, MyCobot-280, SCHUNK EGK-40, Intel RealSense D415, RealSense D435, Orbbec-336L, OAK-D, camera calibration, 3D printing, circuit soldering